

Product stationary laws at capacity two

A disproof of Conjecture 00000001251

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Abstract

On the three-site cycle, capacity-two partial exclusion has a full-support product stationary law with one-site distribution $\text{Bin}(2, 1/2)$. For the alternative convention of unit-rate legal jumps, the product of three uniform laws on $\{0, 1, 2\}$ is stationary. Both examples contradict the assertion that product stationarity requires capacity one.

1 A nondegenerate three-site example

The source defines k -exclusion by the capacity k at each site and asserts that product stationary states occur only at $k = 1$. Consider the three-cycle with sites $0, 1, 2$, so every pair of distinct sites is adjacent. Its state space is $\Omega = \{0, 1, 2\}^3$. A legal jump $i \rightarrow j$ decreases η_i by one and increases η_j by one, requiring $\eta_i > 0$ and $\eta_j < 2$. Every other coordinate remains unchanged.

First use the partial-exclusion rates

$$c_{ij}(\eta) = \eta_i(2 - \eta_j), \quad i \neq j.$$

The off-diagonal generator $Q(\eta, \xi)$ sums these rates over the legal jumps producing ξ ; its diagonal is minus the total exit rate. Define

$$u(0) = \frac{1}{4}, \quad u(1) = \frac{1}{2}, \quad u(2) = \frac{1}{4}, \quad \pi(\eta) = u(\eta_0)u(\eta_1)u(\eta_2).$$

All 27 states have positive mass, and their masses sum to one.

Theorem 1. *The law π is a product stationary distribution for this capacity-two process. Moreover,*

$$\mathbb{E}_\pi \eta_i = 1, \quad \mathbb{E}_\pi(\eta_0 \eta_1 \eta_2) = 1, \quad \mathbb{E}_\pi(\eta_0 \eta_1 \eta_2) - \prod_{i=0}^2 \mathbb{E}_\pi \eta_i = 0.$$

Proof. For a legal jump, write $a = \eta_i \in \{1, 2\}$ and $b = \eta_j \in \{0, 1\}$. The destination has coordinates $a - 1, b + 1$. The binomial weights satisfy

$$u(a)u(b)a(2 - b) = u(a - 1)u(b + 1)(b + 1)(3 - a).$$

Multiplying by the unaffected site's weight gives detailed balance between this jump and its reverse. Summing over jumps yields $\pi(\eta)Q(\eta, \xi) = \pi(\xi)Q(\xi, \eta)$. Thus

$$\sum_{\eta \in \Omega} \pi(\eta)Q(\eta, \xi) = \pi(\xi) \sum_{\eta \in \Omega} Q(\xi, \eta) = 0.$$

For a finite continuous-time chain, this stationarity equation implies $\pi e^{tQ} = \pi$ for every $t \geq 0$, by the exponential power series. The three coordinates are independent with mean one, giving all the stated correlation identities. $\square$

2 The unit-rate convention also has a counterexample

Another common k -exclusion convention assigns rate one to each legal directed jump. Keep exactly the same state space and legal moves, and call its generator Q_U . Each legal jump has a legal reverse of the same rate, so $Q_U(\eta, \xi) = Q_U(\xi, \eta)$. Consequently the full-support law

$$\pi_U(\eta) = \frac{1}{27} = \prod_{i=0}^2 \frac{1}{3}$$

satisfies detailed balance and $\pi_U Q_U = 0$. It is the product of three uniform laws on $\{0, 1, 2\}$, with capacity $2 \neq 1$.

Both examples are genuine particle-moving processes. For example, $(1, 0, 0) \rightarrow (0, 1, 0)$ has rate two in the first process and rate one in the second. Neither law is a point mass. Total particle number is conserved; these are stationary grand-canonical product laws, mixing the invariant particle-number sectors. The source does not require a fixed total particle number or an extremal stationary law. Full independence also removes every mixed three-site correlation deficit that measures failure of factorization.

3 Formal verification and scope

Lean uses the finite state space `Fin 3 × Fin 3 × Fin 3`, with actual coordinate occupations, legal single-particle moves, both rate functions, and both generators. Exact finite computations verify detailed balance for all pairs of states and both stationarity equations.

- `capacity_two`, `weighted_balance`, and `generator_row_sum` verify admissibility and balance.
- `occupation_marginals` and `product_formula` identify the normalized binomial product law. The three-point calculation is `three_point_deficit_zero`.
- `probabilityMass` and `uniformProbabilityMass` are actual Mathlib probability mass functions, with full support.
- `probabilityMass_stationary` and `uniformProbabilityMass_stationary` prove their stationarity equations against the corresponding generators after conversion to real probabilities.

The finite-chain stationarity criterion used is exactly $\pi Q = 0$. Lean verifies that criterion and the generator conditions; it does not construct a separate space of continuous-time sample paths. All checked results use only Lean’s standard logical axioms. No unproved placeholders, custom axioms, native evaluation, or external numerical certificates are used.

Reproduction. In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

Lean 4.19.0 and an exact Mathlib revision are pinned. Rebuild this PDF with `tectonic main.tex`. The exact bilingual source and its provenance are supplied as separate files.

Sources. Conjecture 00000001251; Mathlib c44e0c8ee63c.